MPCA LAB SUBMISSION-6

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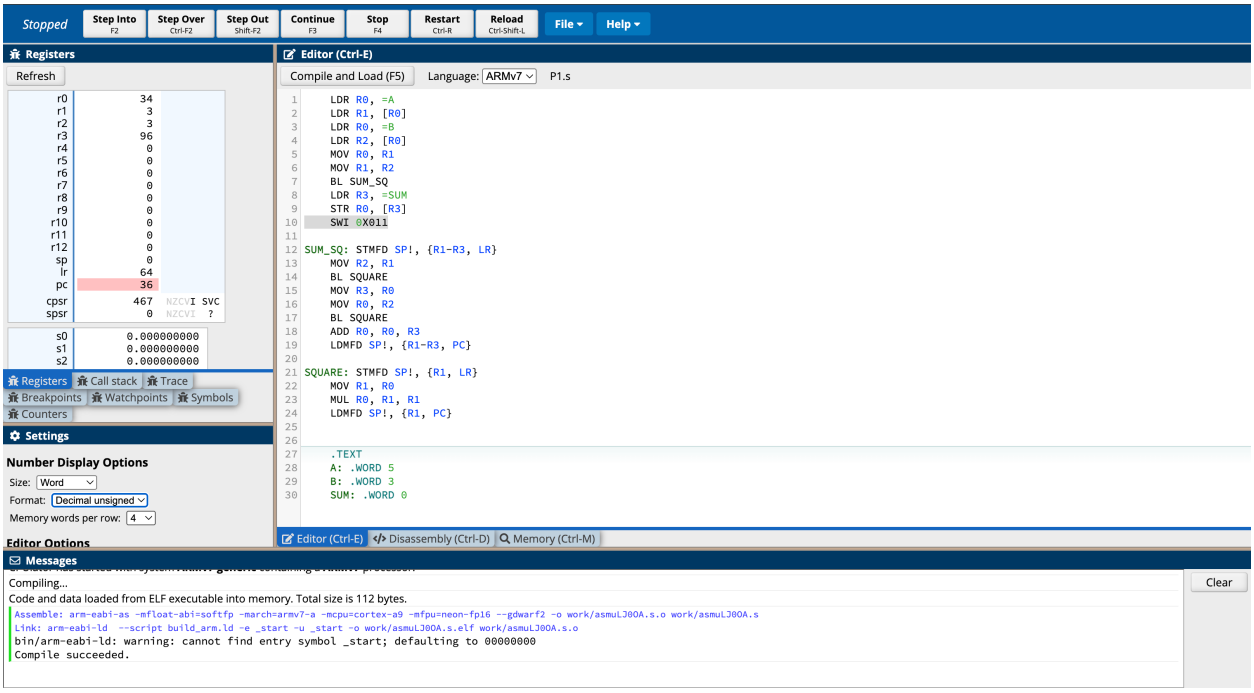

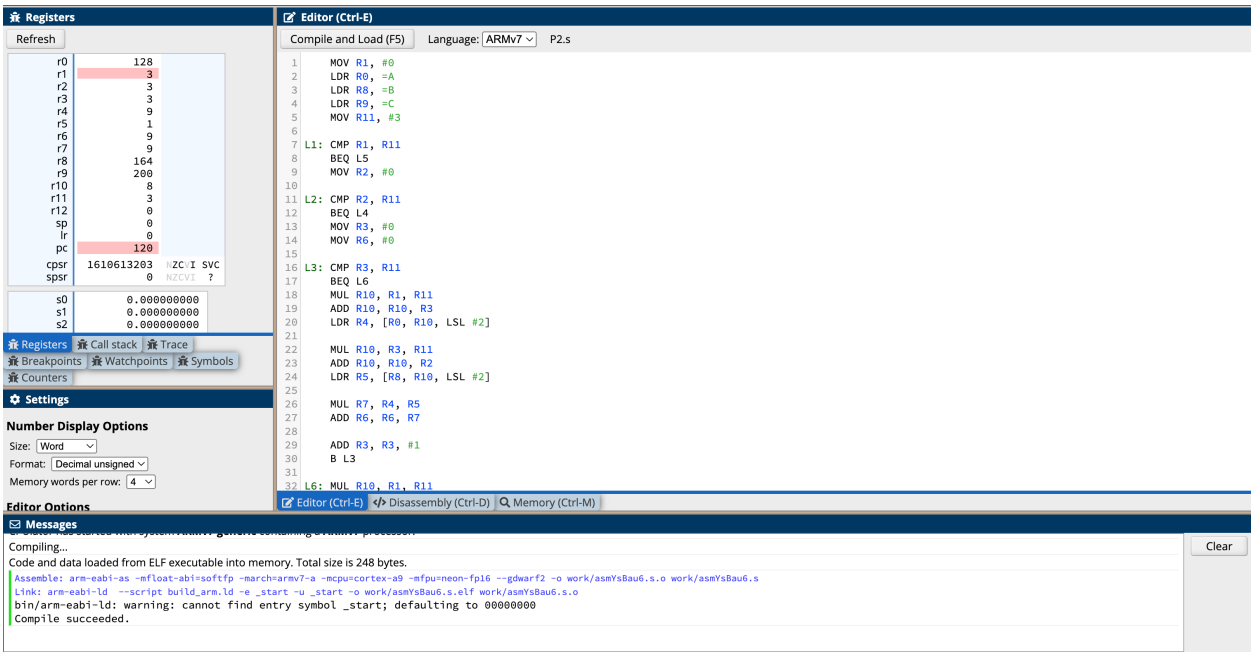

Stopped | Step Into F2 | Step Over Ctrl-F2 | Step Out Shift-F2 | Continue F3 | Stop F4 | Restart Ctrl-R | Reload Ctrl-Shift-L | File | Help

Registers

Refresh

r0	0
r1	64
r2	0
r3	17
r4	3
r5	0
r6	0
r7	0
r8	0
r9	0
r10	0
r11	0
r12	0
sp	0
lr	0
pc	40
cpsr	1610613203 NZC\I SVC
spsr	0 NZCVI ?

s0	0.000000000
s1	0.000000000
s2	0.000000000

Registers | Call stack | Trace | Breakpoints | Watchpoints | Symbols | Counters

Settings

Number Display Options

Size: Word

Format: Decimal unsigned

Memory words per row: 4

Editor Options

Editor (Ctrl-E)

Compile and Load (F5) Language: ARMv7 P3.s

```
    LDR R1, =A
    MOV R2, #5
    MOV R3, #2
    MOV R4, #3
L1: CMP R2, #0
    BEQ L2
    STR R3, [R1], #4
    ADD R3, R3, R4
    SUB R2, R2, #1
    B L1
L2: SWI 0X011
    .TEXT
    A: .WORD 0, 0, 0, 0, 0
```

Editor (Ctrl-E) | Disassembly (Ctrl-D) | Memory (Ctrl-M)

Messages

Clear

```
Compiling...
Code and data loaded from ELF executable into memory. Total size is 72 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmOAjZSv.s.o work/asmOAjZSv.s
Link: arm-eabi-ld  --script build_arm.ld -e _start -u _start -o work/asmOAjZSv.s.elf work/asmOAjZSv.s.o
bin/arm-eabi-ld: warning: cannot find entry symbol _start; defaulting to 00000000
Compile succeeded.
```

Registers

Refresh

r0	0
r1	76
r2	0
r3	10
r4	1
r5	0
r6	10
r7	80
r8	0
r9	0
r10	0
r11	0
r12	0
sp	0
lr	0
pc	72
cpsr	1610613203 NZC\I SVC
spsr	0 NZCVI ?

s0	0.000000000
s1	0.000000000
s2	0.000000000

Registers | Call stack | Trace | Breakpoints | Watchpoints | Symbols | Counters

Settings

Number Display Options

Size: Word

Format: Decimal unsigned

Memory words per row: 4

Editor Options

Editor (Ctrl-E)

Compile and Load (F5) Language: ARMv7 P4.s

```
    LDR R1, =NUM
    LDR R2, [R1]
    MOV R3, #0
LP1: CMP R2, #0
    BEQ END
    MOV R4, R2
    MOV R5, #0
    MOV R6, #10
DIV: CMP R4, R6
    BLT REM
    SUB R4, R4, R6
    ADD R5, R5, #1
    B DIV
REM: ADD R3, R3, R4
    MOV R2, R5
    B LP1
END: LDR R7, =SUM
    STR R3, [R7]
    SWI 0x011
    .TEXT
    NUM: .WORD 1234
    SUM: .WORD 0
```

Editor (Ctrl-E) | Disassembly (Ctrl-D) | Memory (Ctrl-M)

Messages

Clear

```
Compiling...
Code and data loaded from ELF executable into memory. Total size is 96 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmMoAUtd.s.o work/asmMoAUtd.s
Link: arm-eabi-ld  --script build_arm.ld -e _start -u _start -o work/asmMoAUtd.s.elf work/asmMoAUtd.s.o
bin/arm-eabi-ld: warning: cannot find entry symbol _start; defaulting to 00000000
Compile succeeded.
```

Registers

Refresh

r0	00000034	
r1	00000004	
r2	00000000	
r3	00000048	
r4	0000004f	
r5	00000000	
r6	00000000	
r7	00000000	
r8	00000000	
r9	00000000	
r10	00000000	
r11	00000000	
r12	00000000	
sp	00000000	
lr	00000000	
pc	00000030	
cpsr	600001d3	NZCVI SVC
spsr	00000000	NZCVI ?

s0	00000000
s1	00000000
s2	00000000

Registers | Call stack | Trace | Breakpoints | Watchpoints | Symbols | Counters

Settings

Number Display Options

Size: Word

Format: Hexadecimal

Memory words per row: 4

Editor Options

Editor (Ctrl-E)

Compile and Load (F5) Language: ARMv7 P5-a.s

```
    LDR R0, =A
    MOV R1, #0

L1: LDRB R2, [R0, R1]
    CMP R2, #0
    BEQ L2
    ADD R1, R1, #1
    B L1

L2: SUB R1, R1, #1
    LDRB R3, [R0]
    LDRB R4, [R0, R1]
    STRB R4, [R0]
    STRB R3, [R0, R1]

    SWI 0X011

    .TEXT
    A: .ASCIZ "HELLO"
```

Editor (Ctrl-E) | Disassembly (Ctrl-D) | Memory (Ctrl-M)

Messages

Clear

Compiling...

Code and data loaded from ELF executable into memory. Total size is 64 bytes.

```
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmaD6Q6h.s.o work/asmaD6Q6h.s
Link: arm-eabi-ld  --script build_arm.ld -e _start -u _start -o work/asmaD6Q6h.s.elf work/asmaD6Q6h.s.o
bin/arm-eabi-ld: warning: cannot find entry symbol _start; defaulting to 00000000
Compile succeeded.
```

Registers

Refresh

r0	44	
r1	5	
r2	9	
r3	20	
r4	9	
r5	29	
r6	48	
r7	0	
r8	0	
r9	0	
r10	0	
r11	0	
r12	0	
sp	0	
lr	0	
pc	36	
cpsr	467	NZCVI SVC
spsr	0	NZCVI ?

s0	0.000000000
s1	0.000000000
s2	0.000000000

Registers | Call stack | Trace | Breakpoints | Watchpoints | Symbols | Counters

Settings

Number Display Options

Size: Word

Format: Decimal unsigned

Memory words per row: 4

Editor Options

Editor (Ctrl-E)

Compile and Load (F5) Language: ARMv7 P5-b.s

```
    LDR R0, =A
    LDR R1, [R0]

    LDR R0, =B
    LDR R2, [R0]

    MOV R3, R1, LSL #2
    AND R4, R2, #15
    ORR R5, R3, R4

    LDR R6, =Z
    STR R5, [R6]

    SWI 0X011

    .TEXT
    A: .WORD 5
    B: .WORD 9
    Z: .WORD 0
```

Editor (Ctrl-E) | Disassembly (Ctrl-D) | Memory (Ctrl-M)

Messages

Clear

Compiling...

Code and data loaded from ELF executable into memory. Total size is 64 bytes.

```
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asm20I8Qo.s.o work/asm20I8Qo.s
Link: arm-eabi-ld  --script build_arm.ld -e _start -u _start -o work/asm20I8Qo.s.elf work/asm20I8Qo.s.o
bin/arm-eabi-ld: warning: cannot find entry symbol _start; defaulting to 00000000
Compile succeeded.
```

Stopped | Step Into F2 | Step Over Ctrl-F2 | Step Out Shift-F2 | Continue F3 | Stop F4 | Restart Ctrl-R | Reload Ctrl-Shift-L | File | Help

Registers

Refresh

r0	136	
r1	0	
r2	5	
r3	147	
r4	135	
r5	0	
r6	0	
r7	0	
r8	0	
r9	0	
r10	0	
r11	0	
r12	0	
sp	4294967268	
lr	48	
pc	128	
cpsr	1610613203	NZCVI SVC
spsr	0	NZCVI ?

s0	0.000000000
s1	0.000000000
s2	0.000000000

Registers | Call stack | Trace | Breakpoints | Watchpoints | Symbols | Counters

Settings

Number Display Options

Size: Word

Format: Decimal unsigned

Memory words per row: 4

Editor Options

Editor (Ctrl-E)

Compile and Load (F5) | Language: ARMv7 | P6.s

```
    LDR R0, =A
    BL STRING_PROC
    LDR R0, =REV
    SWI 0x02
    SWI 0X011

STRING_PROC: STMFD SP!, {R1, R2, LR}
    MOV R1, R0
    BL LENGTH
    MOV R2, R0
    MOV R0, R1
    MOV R1, R2
    BL REVERSE
    LDMFD SP!, {R1, R2, PC}

LENGTH: MOV R2, #0
LEN_LOOP: LDRB R3, [R0], #1
    CMP R3, #0
    BEQ LEN_END
    ADD R2, R2, #1
    B LEN_LOOP
LEN_END: MOV R0, R2
    BX LR

REVERSE: STMFD SP!, {R3 - R5, LR}
    LDR R3, =REV
    ADD R4, R0, R1
    SUB R4, R4, #1
REV_LOOP: CMP R1, #0
    BEQ REV_END
    LDRB R5, [R4], #-1
    STRB R5, [R3], #1
```

Editor (Ctrl-E) | Disassembly (Ctrl-D) | Memory (Ctrl-M)

Messages

Clear

Compiling...

Code and data loaded from ELF executable into memory. Total size is 176 bytes.

```
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asm3mXzua.s.o work/asm3mXzua.s
Link: arm-eabi-ld  --script build_arm.ld -e _start -u _start -o work/asm3mXzua.s.elf work/asm3mXzua.s.o
bin/arm-eabi-ld: warning: cannot find entry symbol _start; defaulting to 00000000
Compile succeeded.
```

```
    MOV R2, R0
    MOV R0, R1
    MOV R1, R2
    BL REVERSE
    LDMFD SP!, {R1, R2, PC}

LENGTH: MOV R2, #0
LEN_LOOP: LDRB R3, [R0], #1
    CMP R3, #0
    BEQ LEN_END
    ADD R2, R2, #1
    B LEN_LOOP
LEN_END: MOV R0, R2
    BX LR

REVERSE: STMFD SP!, {R3 - R5, LR}
    LDR R3, =REV
    ADD R4, R0, R1
    SUB R4, R4, #1
REV_LOOP: CMP R1, #0
    BEQ REV_END
    LDRB R5, [R4], #-1
    STRB R5, [R3], #1
    SUB R1, R1, #1
    B REV_LOOP
REV_END: MOV R5, #0
    STRB R5, [R3]
    LDMFD SP!, {R3 - R5, PC}

    .TEXT
    A: .ASCIZ "Hello"
    REV: .WORD 0, 0, 0, 0, 0
```